

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Linear Circuit Analysis	Course Code:	EE1001
	Program:	Electrical Engineering	Semester:	Spring 2023
	Duration:	1 Hour	Total Marks:	30
	Exam Date:	10 April 2023	Weight:	15
	Section:	All	Page(s):	2
	Exam Type:	Sessional - 2		

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- Instruction/Notes:**
1. Attempt all questions on the provided answer sheet.
 2. Show all steps and box all the final answers with proper units.
 3. Attempt all parts of a question together.

CLO 02

Question No. 1

Marks: 10

For the circuit shown below, use node-voltage method to find the power of the dependent source and 1A independent current source.

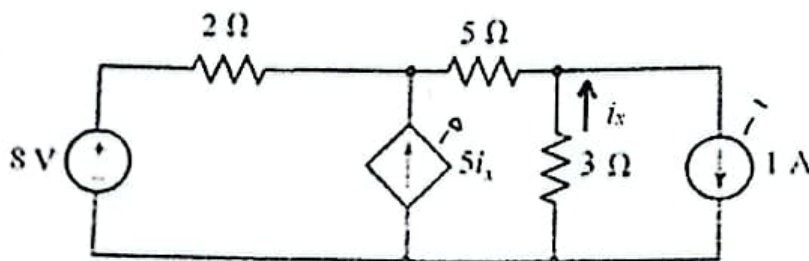


Figure 1. Circuit containing dependent source

Question No. 2

Marks: 10

Use repeated source transformations to find the value of v_x .

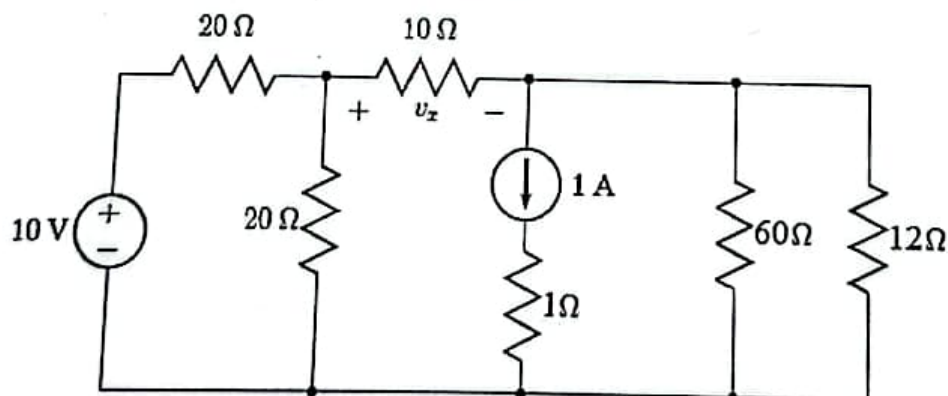


Figure 2. Circuit for Source Transformation

Question No. 3

Marks: 10

For the circuit shown below,

- Calculate the Thévenin equivalent voltage with respect to the terminals a, b using **mesh current method**
- Calculate the Thevenin Equivalent resistance at the terminals a, b using any suitable method.

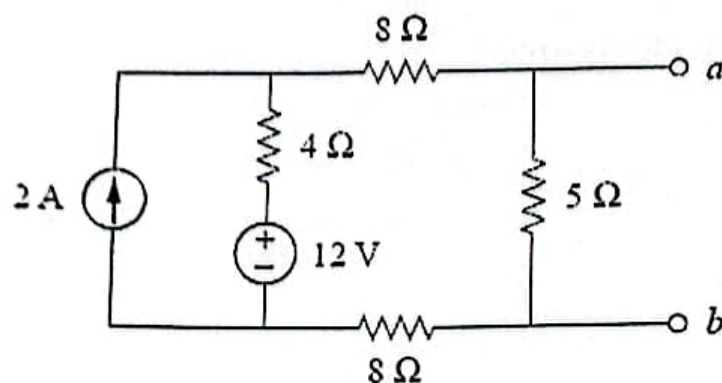


Figure 3. Circuit for finding Thevenin Equivalent

	Course Name:	Linear Circuit Analysis	Course Code:	EE1001
	Program:	BS(Electrical Engineering)	Semester:	Spring 2023
	Duration:	60 minutes	Total Marks:	30
	Paper Date:	28-February-2023	Weightage:	15%
	Section:	All	Page(s):	2
	Exam Type:	MID - I		

Student Name: Muhammad Pervez Mayo Roll No. 221-7739 Section: _____

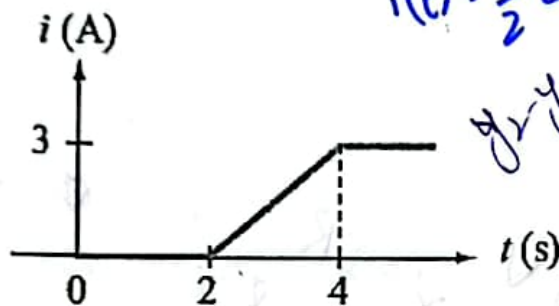
- Instruction/Notes:
1. Circuits must be labeled properly, methods/techniques must be indicated and formulas must be written prior to inserting values.
 2. Show all steps.
 3. Attempt all parts of a question together.

Q.1

[10 marks]

[CLO 01]

The current $i(t)$ passing through a 10 mH inductor is given by the waveform shown in Figure 1 below:



$$i(t) = \frac{1}{2} L \int_0^t v dt + i(0)$$

$y_2 - y_1 = m(x_2 - x_1)$

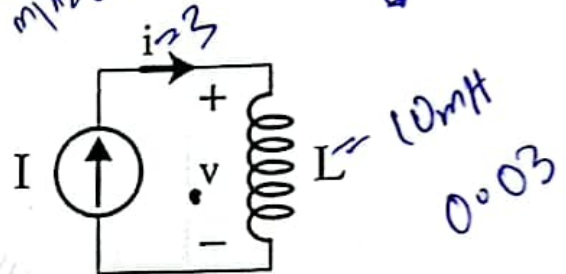


Figure 1. Waveform and Circuit for the current through the inductor

- Determine the piece-wise equation of the current from the graph.
- Determine corresponding voltage across the inductor and plot it.

Q.2 For the circuit shown in Figure 2 below: [10 marks]

[CLO 02]

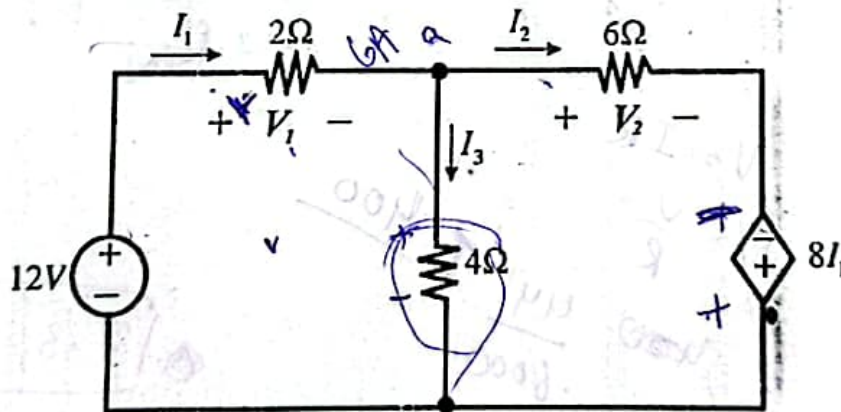
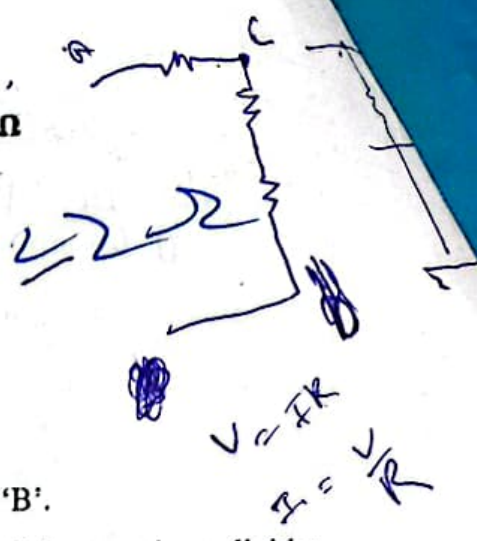


Figure 2. Circuit with Dependent Source

- Determine the voltages V_1 , V_2 and currents I_1 , I_2 using Kirchhoff's laws and Ohm's Law.
- Determine the power associated with the dependent voltage source using passive sign convention.

1CLO 0247


$$V = IR$$
$$I = \frac{V}{R}$$

- $$\begin{aligned} &= I R_1 + I R_2 + I R_3 \\ &= I V_1 + I V_2 + I V_3 \\ &= I (V_1 + V_2 + V_3) \end{aligned}$$

$P = \sqrt{I^2 R}$

$$V = \frac{R_g \times i_T}{R_{eq}}$$

$$I = \frac{V}{R}$$
$$= \frac{44}{6000}$$
$$I = 400$$

$$7.3 \times 10^{-3} \checkmark$$

$$I = \frac{V}{R}$$

④. 7.3 mV